Anticipated Theoretical & Methodological Questions

BrainKAN Interpretability Benchmark

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This document addresses the key theoretical and statistical questions regarding the BrainKAN Interpretability Benchmark. It clarifies the exact boundaries between conditional optimization artifacts, reference-domain dependencies, and population-level scientific claims.

Q1: Why does null calibration not equate to identifying biological truth?

Answer: It strictly does **not** identify in-vivo biological mechanisms.
What the null calibration isolates is the *excess structural deviation relative to the empirical null expectation under the specific optimization procedure*. The null model (trained on label-permuted data) preserves the graph topology and the model's architectural flexibility while destroying task supervision. A residual metric indicates whether the learned geometry is associated with the training label. It proves how the computational model routes information in-silico, not that the biological brain executes this exact mathematical function locally.

Q2: Why can a single seed (e.g., Seed 42) achieve $p = 0.006$, yet fail to generalize across seeds?

Answer: Because of an **Estimand Mismatch** (Conditional significance without population-level generalization).
Seed 42's result is not a computational mistake or a statistical false positive. It validly tests the conditional null hypothesis:

$$H_0^{(s=42)} : \Delta_{s=42} = 0 \tag{1}$$

For that specific initialization basin, the real-label model converged to higher curvature than its paired null models. However, the scientific hypothesis of interest is the *marginal population estimand*:

$$H_0 : \mathbb{E}_{s\sim\mathcal{S}}[\Delta_s] \leq 0 \tag{2}$$

When marginalizing across 10 independent initialization seeds, 9 out of 10 seeds yield negative differences ($\bar{\Delta} = -0.00058$, $p = 0.9513$). Single-seed inference does not generalize across optimization randomness.

Q3: Why is nonlinearity measurement reference-domain dependent?

Answer: Because curvature and Deviation-From-Linearity (DFL) are operators conditioned on the evaluation support.
As shown in Synthetic Calibration (Exp 2-B), evaluating the exact same static mathematical function over shifted input domains causes apparent DFL to jump by $+0.1195$—an artifact two orders of magnitude larger than actual model differences. Unless evaluation is strictly bounded to the common support ($\text{supp}(X_0) \cap \text{supp}(X_1)$), distribution shifts will create false non-linear interpretations.

Q4: What is the true invariant learned by BrainKAN?

Answer: The intrinsic baseline curvature budget ($T \approx 0.179 \pm 0.001$).
Regardless of initialization seeds, cross-validation folds, or label permutations, the edge B-splines consistently allocate a substantial degree of functional nonlinearity to route input node features. The model's capacity naturally utilizes curved representations, but task supervision does not drive excess curvature beyond this baseline.

Q5: What is the recommended standard for future interpretable KAN research?

Answer: The Seed-Marginalized Permutation Testing (SMPT) with Common Support protocol.
Future studies evaluating spline nonlinearity must: (1) enforce strict common-support domain bounds to prevent extrapolation artifacts, (2) compute paired permutation nulls, and (3) report the seed-marginalized effect $\mathbb{E}_s[\Delta_s]$ across multiple random initializations.